DATA JOB MARKET ANALYSIS

Power BI Dashboard & Business Insights Report

Dataset records	Companies	Industries	Average salary
742	343	60	USD 101.48K*

Report basis: Power BI dashboard screenshots and project details shared during the walkthrough. This report distinguishes visible dashboard findings from items that still require source-data validation.

Prepared for portfolio and project submission

1. Executive Summary

This project analyzes a dataset of 742 job postings to explore the distribution of job opportunities across locations, job roles, industries and companies, together with salary, education, company-rating and technical-skill indicators. The dataset was prepared in Power Query and analyzed in Power BI using DAX measures and interactive visuals.

The dashboard reports 343 distinct companies and 60 industries. The average salary measure is 101.48, interpreted in the dashboard as USD thousands (approximately USD 101.48K). This unit interpretation should be confirmed against the source field before external publication.

Headline metrics

Metric	Dashboard value	Description
Job postings	742	Rows/job postings represented in the dashboard
Companies	343	Distinct company names counted
Industries	60	Distinct industry values counted
Average salary	101.48	Average salary measure; assumed USD thousands

Scope: These figures describe the records in this dataset only. They should not be interpreted as a complete or live representation of the entire job market.

2. Project Objectives

• Summarize job postings, companies and industries.
• Identify locations with the highest posting counts.
• Compare demand across job roles and industries.
• Explore salary patterns by job role and education category.
• Compare company ratings and selected technical-skill indicators.
• Present findings in an interactive Power BI dashboard.

3. Dataset Overview

The Power Query preview showed 742 rows and 42 columns before one column was removed; the loaded model displayed 41 columns. Visible fields included job title, salary estimate, company name, job location, headquarters, company size, founded year, industry, sector, revenue, competitors, rating, age, education and technology/skill indicators. These fields support comparisons of job postings and company attributes.

The source URL, collection date, exact geographic coverage and complete data dictionary were not visible in the supplied evidence and are not asserted in this report.

4. Data Preparation in Power Query

The CSV data was imported into Power Query, the first row was promoted to headers, and data types were assigned. One unnecessary column was removed, leaving 41 columns in the loaded model. Placeholder values of -1 were replaced with null for fields including Founded and Age, with similar cleanup applied to selected categorical fields. The displayed column profiles showed no conversion errors at the time of review.

Cleaning principles applied

• Treat -1 placeholders as unavailable values rather than real measurements.
• Preserve valid founding-year and age values.
• Keep numeric fields numeric and categorical fields as text.
• Retain the 742 records while removing an unnecessary column.

Validation still required: The screenshots do not prove that every missing value, duplicate or placeholder across all 41 columns has been fully audited. Before submission, verify row counts, duplicate IDs, missing-value counts and final data

types.

5. Power BI Measures

The dashboard uses DAX measures for headline KPIs and chart values. The measures created or discussed during the project include:

Measure	Purpose
Total Job Openings	Counts rows in the data table.
Average Salary	Calculates the average of Avg_SalaryK.
Minimum Salary	Returns the minimum Lower_Salary value.
Maximum Salary	Returns the maximum Upper_Salary value.
Total Companies	Counts distinct company names.
Total Industries	Counts distinct industry values.
Average Rating	Calculates the average company rating.
Python Jobs, SQL Jobs, Excel Jobs, AWS Jobs, Spark Jobs	Sums the corresponding skill-indicator columns.

Salary values are presented as USD thousands (K) in the dashboard. Confirm the salary parsing, source currency and units before using this metric for formal decisions.

6. Key Findings

6.1 Geographic distribution

California has the largest visible count with 152 postings, followed by Massachusetts with 103 and New York with 72. Virginia (41), Illinois (40), Maryland (35), Pennsylvania (33) and Texas (28) also appear among the leading states shown in the dashboard.

State	Postings
California (CA)	152
Massachusetts (MA)	103
New York (NY)	72
Virginia (VA)	41
Illinois (IL)	40
Maryland (MD)	35
Pennsylvania (PA)	33
Texas (TX)	28

Interpretation: Job postings in this dataset are concentrated in a small number of states, with California clearly ahead. This describes the dataset distribution and does not explain the causes of the concentration.

6.2 Job-role demand

Data Scientist is the largest visible job-title group with 251 postings. Other Scientist has 111, Data Engineer has 77, Analyst has 59 and Machine Learning Engineer has 18 postings.

Job role	Postings
Data Scientist	251
Other Scientist	111
Data Engineer	77
Analyst	59
Machine Learning Engineer	18

6.3 Industry demand

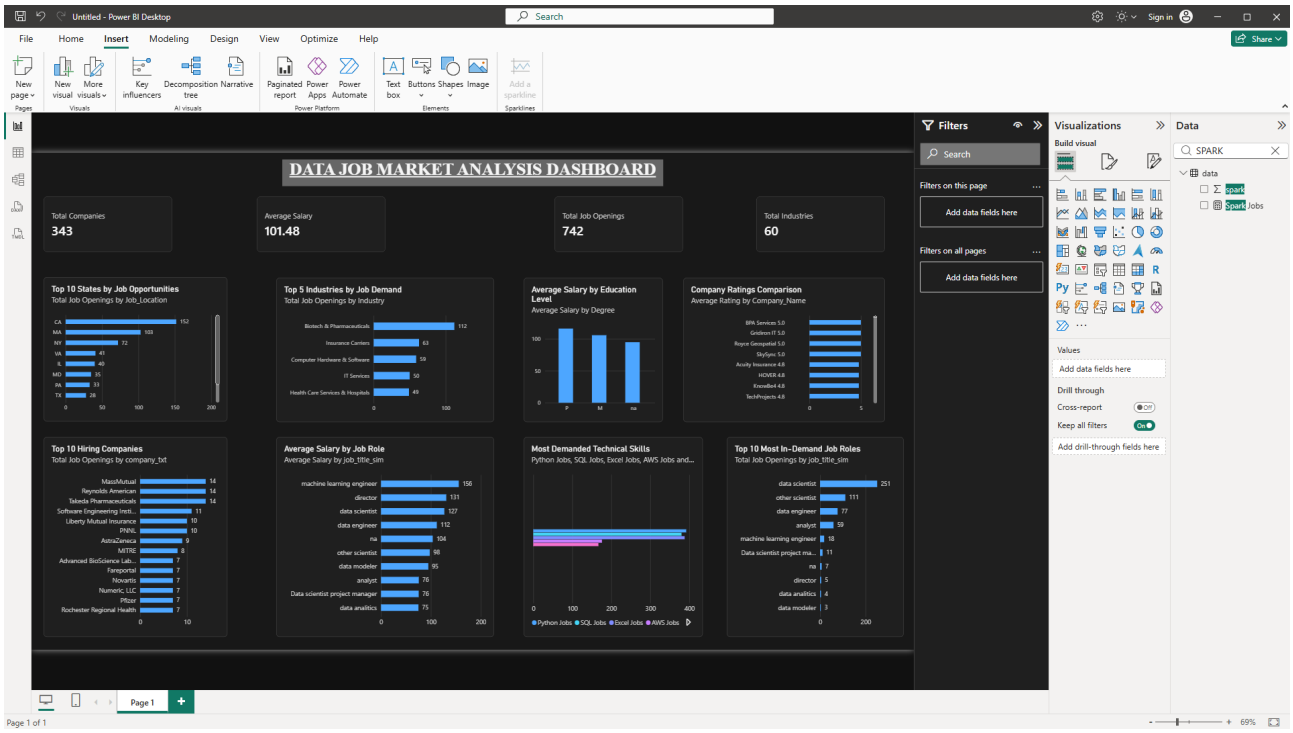
Biotech & Pharmaceuticals leads the visible industry chart with 112 postings. Insurance Carriers follows with 63, Computer Hardware & Software with 59, IT Services with 50, and Health Care Services & Hospitals with 49. These results show strong representation from life sciences, insurance, technology and healthcare in this dataset.

6.4 Salary, education, ratings and skills

The dashboard includes average salary by job role, average salary by education category, company ratings and indicators for Python, SQL, Excel, AWS and Spark. The visible education categories include abbreviated labels (P, M and na), and the skill chart does not show readable numeric labels in the supplied screenshot. Therefore, this report does not claim a definitive salary premium by education or a numerical ranking of skills. These views should be validated before stronger conclusions are drawn.

7. Dashboard Snapshot

The screenshot below records the Power BI dashboard at the time of review. It includes KPI cards, geographic and industry charts, hiring-company analysis, salary visuals, company ratings, skill indicators and job-role demand.



8. Business Recommendations

1. **Prioritize high-volume locations.** Use the state distribution to guide further research into opportunities in California, Massachusetts and New York. Compare remote, hybrid and on-site roles.
2. **Build role-specific learning plans.** Since Data Scientist and Data Engineer roles are prominent, compare the skills requested in individual postings and align learning plans accordingly.
3. **Investigate industry specialization.** Explore skills, salary ranges and education expectations in biotech, insurance, technology and healthcare postings.
4. **Use salary ranges as well as averages.** Compare lower and upper salary estimates by role and location to understand variation.
5. **Validate employer and skill comparisons.** Standardize company names, consider rating sample sizes and verify skill indicators before ranking employers or technologies.

9. Limitations

- The analysis covers 742 dataset records, not the entire job market.
- The collection date and source provenance were not available in the evidence reviewed.
- Salary estimates, currency and units need confirmation against the source data.
- The Top N company chart can display more than ten companies when values are tied at the cutoff.
- Education labels are abbreviated and should be mapped to readable categories.
- Skill comparisons and company-rating sample sizes require validation.
- Missing values and duplicates should be audited across the final dataset.

10. Conclusion

The Power BI dashboard provides an exploratory view of 742 job postings across 343 companies and 60 industries. The visible results show that California has the largest state-level posting count, Data Scientist is the most common displayed role, and Biotech & Pharmaceuticals is the leading displayed industry. The dashboard also brings salary, education, company-rating and technical-skill indicators into one report. After validating source fields, cleaning category labels and

checking calculations, this project can serve as a portfolio example of data preparation, DAX analysis and dashboard storytelling.

Appendix: Final Submission Checklist

- ■ Save the final Power BI file (.pbix).
- ■ Keep the original CSV files with the project source materials.
- ■ Verify all DAX measures and confirm salary units.
- ■ Check duplicate IDs, missing values and data types in Power Query.
- ■ Replace abbreviated education labels with readable descriptions where possible.
- ■ Ensure chart titles, axis labels and data labels are readable.
- ■ Export the final dashboard page as a PDF or image.
- ■ Add the report, PBIX file, dataset documentation and screenshots to the repository.

Report basis: Power BI screenshots and metrics shared during the project walkthrough. Values not visible or independently verifiable from those materials have not been invented.